

1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

ANSWER:-

Feature	DDR3	DDR4	DDR5
Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5V (1.35V for DDR3L)	1.2V	1.1V
Typical Use Cases	Older PCs, Basic Office Work	Gaming, Productivity, General Computing	High-end Gaming, AI, Video Editing, Workstations
Release Year	2007	2014	2020
Maximum Module Capacity	Up to 16 GB	Up to 64 GB	64 GB and above
Power Consumption	Higher	Lower than DDR3	Lowest
Compatibility	DDR3 Motherboards Only	DDR4 Motherboards Only	DDR5 Motherboards Only

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM Module. Reinstall the RAM module and verify that your system boots up successfully.

Hint : Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

ANSWER:-

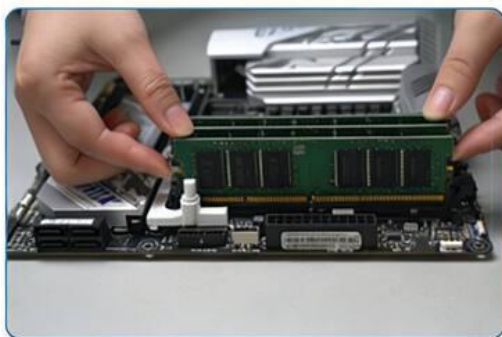
- RAM Module Before and After Reinstallation



1. Open the computer case, check the memory slot position, discharge static electricity from your hands, and clean the dust in the slot



2. Open the latches on both sides of the memory slot



3. Align with the memory key notch, if it doesn't fit, rotate the memory module



4. After aligning with the key notch, press down on the memory module until a "click" sound is heard, and check if the golden fingers are exposed. If not, the installation is successful

Objective:

- To remove the RAM module safely, reinstall it correctly, and verify that the computer boots successfully after reassembly.

Hardware Specifications:

- Motherboard: ASUS ROG Maximus Z790 Formula
- Memory: Corsair DDR5 16GB 6000MHz RAM

Step-by-Step Procedure:

✓ RAM Identification (Panel 1): The computer was turned off and disconnected from the power supply. The RAM module was located on the motherboard before removal.

✓ RAM Removal (Panel 2): The retaining clips on both sides of the RAM slot were pressed outward, and the RAM module was carefully removed.

✓ RAM Reinstallation (Panel 3): The RAM module was aligned with the slot notch and inserted firmly until both retaining clips locked into place.

✓ System Verification (Panel 4): The computer was powered on after reassembly. The system booted successfully, and the installed RAM was detected correctly in Windows Task Manager.

Result: The RAM module was successfully reinstalled, and the system booted normally without any memory errors.

3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

ANSWER:-

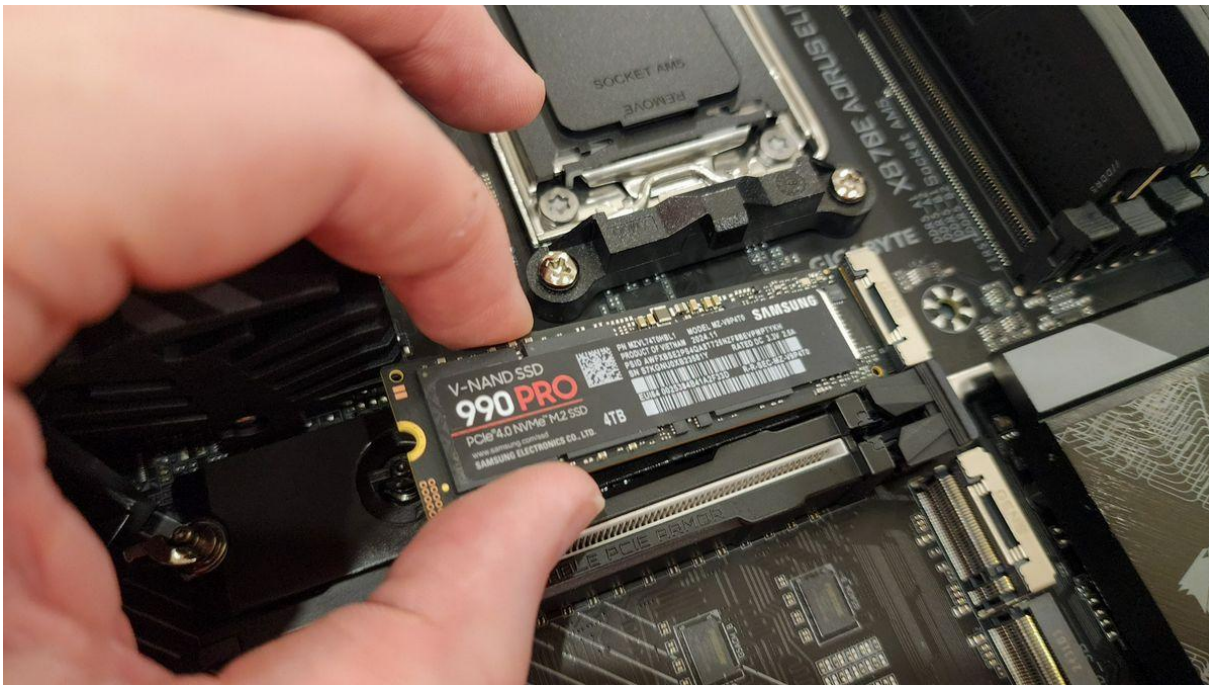
Feature	HDD	SSD (SATA)	NVMe SSD
Speed	Slow (80–160 MB/s)	Fast (500–550 MB/s)	Very Fast (2000–7000+ MB/s)
Technology	Magnetic Disk	NAND Flash Memory	NAND Flash Memory over PCIe
Best Daily App/Use Case	Movies, Photos, File Backup	Microsoft Office, Google Chrome, Windows Boot	Adobe Premiere Pro, Photoshop, Modern Games, Visual Studio
Cost	Lowest cost	Moderate cost	Highest cost

4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model. Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

ANSWER:-

Storage Drive Removal and Verification



Objective:

- To remove the storage drive, identify its connector type, reinstall it correctly, and verify that the system detects the drive after reassembly.

Hardware Specifications:

- Motherboard: ASUS ROG Maximus Z790 Formula
- Storage Drive: Samsung 990 Pro 2TB M.2 NVMe SSD

Step-by-Step Procedure:

✓ **Drive Removal (Panel 1):** The computer was powered off and disconnected from the power supply. The M.2 heatsink was removed carefully, and the SSD was taken out from the motherboard.

✓ **Storage Drive Identification (Panel 2):** The storage drive connector was identified as an M.2 NVMe SSD connector.

✓ **Drive Reinstallation (Panel 3):** The SSD was aligned with the M.2 slot and inserted correctly. The retaining screw and heatsink were reinstalled securely.

✓ **System Verification (Panel 4):** After reassembling the computer, it was powered on. Windows successfully detected the SSD, and the drive appeared in Disk Management and File Explorer.

Result: The storage drive was successfully reinstalled, and the system recognized it correctly without any errors.